

EXERCISE 4

JOINS, UNION, FILTERING AND AGGREGATIONS

1. SELECT EmployeeID, firstName, LastName, Department, Salary, ProjectName, Budget, Status, ProjectID
FROM Employees AS A
INNER JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;

Employee ID	First Name	Last Name	Department	Salary	Project Name	Budget	Project ID	Status
1	John	Doe	IT	70 000	AI Development	100 000	101	Completed
2	Alice	Smith	HR	60 000	Employee Training	50 000	102	Ongoing
1	John	Doe	IT	70 000	Cybersecurity Audit	75 000	103	Pending
3	Bob	Johnson	Finance	75 000	Financial Analysis	90 000	104	Ongoing
5	Emma	Wilson	Sales	65 000	Market Expansion	65 000	105	Completed
6	Michael	Clark	Finance	80 000	Risk Management	80 000	106	Pending

2. SELECT EmployeeID, firstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status
FROM Employees AS A
LEFT JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

4. SELECT EmployeeID, firstName, LastName, Department, Salary,
 ProjectID, projectName, Budget, Status
 FROM Employees AS A
 FULL OUTER JOIN Projects AS B
 ON A.EmployeeID = B.EmployeeID;

Employee ID	FirstName	LastName	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

3. SELECT ProjectID, ProjectName, Budget, Status, EmployeeID,
 firstName, LastName, Department, Salary
 FROM Employees AS A
 RIGHT JOIN Projects AS B
 ON A.EmployeeID = B.EmployeeID;

Project ID	Project Name	Budget	Status	Employee ID	FirstName	LastName	Department	Salary
101	AI Development	100 000	Completed	1	John	Doe	IT	70 000
102	Employee Training	50 000	Ongoing	2	Alice	Smith	HR	60 000
103	Cybersecurity Audit	75 000	Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90 000	Completed	3	Bob	Johnson	Finance	75 000
105	Market Expansion	65 000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk Management	80 000	Pending	6	Michael	Clark	Finance	80 000

\$ SELECT city AS location
FROM employees
UNION

SELECT status FROM projects;

location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

6. SELECT city AS location
FROM employees
UNION ALL

SELECT status FROM projects;

location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
Pending
Completed
Pending

7. SELECT EmployeeID, firstName, LastName, Department,
Salary
FROM employees
WHERE Salary > 70 000;

EmployeeID	firstName	LastName	Department	Salary
3	Bob	Johnson	finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	finance	80 000

8. SELECT EmployeeID, first Name, Last Name, Department,
Salary
FROM employees
WHERE Department IN ('IT', 'finance');

EmployeeID	firstName	Department	Salary
1	John	IT	70 000
3	Bob	finance	75 000
4	David	IT	72 000
6	Michael	finance	80 000

9. SELECT projectID, projectName, Budget, Status
FROM projects
WHERE status <> 'Completed';

project ID	projectName	Budget	Status
102	Employee Training	50 000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

10. SELECT projectID, projectName, Budget, status
FROM projects
WHERE Budget > 70000 AND status <> 'Completed';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending.

11. SELECT EmployeeID, firstName, LastName, Department, Salary,
city
FROM employees
WHERE city IN ('New York', 'Toronto')
ORDER BY Salary DESC;

EmployeeID	firstName	City	Salary
6	Michael	New York	80000
3	Bob	Toronto	75000
1.	John	New York	70000

12. SELECT EmployeeID, firstName, LastName, Department,
Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;

EmployeeID	firstName	LastName	Salary
6	Michael	Clark	80000
3	Bob	Johnson	75000
4	David	Brown	72000

13. SELECT Department, SUM (salary) AS TotalSalary
 FROM employees
 GROUP BY Department
 ORDER BY TotalSalary DESC;

Department	TotalSalary
finance	155 000
IT	142 000
HR	60 000
Sales	65 000

14. SELECT City, AVG (salary) AS AveragedSalary
 FROM employees
 GROUP BY City
 HAVING AVG (salary) > 65 000;

City	AveragedSalary
New York	75 000
Toronto	75 000
London	72 000

15. SELECT Department, COUNT (EmployeeID) AS EmployeeCount
 FROM employees
 GROUP BY Department
 HAVING COUNT (EmployeeID) > 1;

Department	EmployeeCount
IT	2
Finance	2

16. SELECT Status, COUNT (project ID) AS project count
 FROM projects
 GROUP BY Status
 HAVING COUNT (project ID) >= 2;

Status	Project count
Completed	2
Ongoing	2
Pending	2

17. SELECT Employee ID, first Name, last Name,
 sum (Budget) AS Total Project Budget
 FROM employees AS A
 JOIN projects AS B
 ON. A. Employee ID = B. Employee ID
 GROUP BY Employee ID, first Name, last Name
 HAVING sum (Budget) > 150 000;

Employee ID	first Name	Total Project Budget
1	John	175 000